ACCT102 Management Accounting

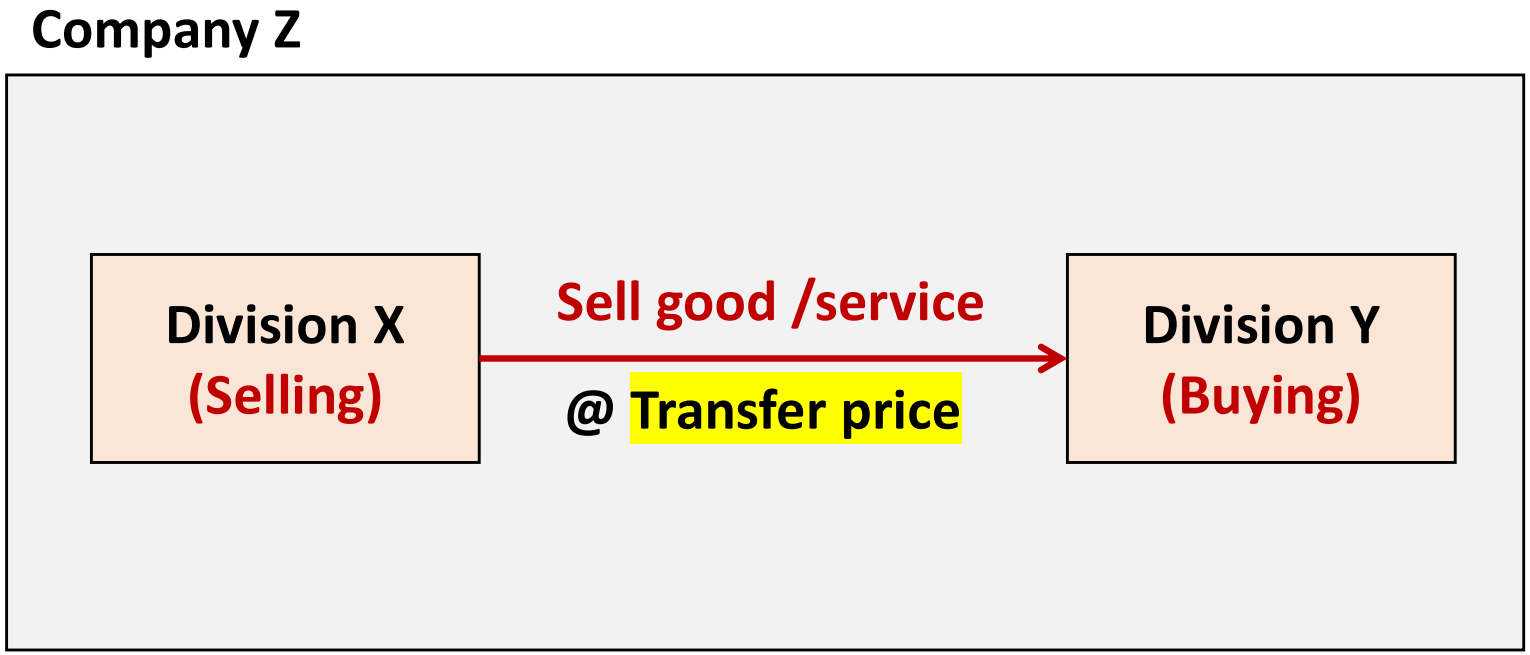
Week 13

Transfer Pricing

Adapted from © McGraw-Hill Education for ACCT102 (BT)

Transfer Pricing

Price charged when one segment of a company provides goods or services to another segment of the same company.



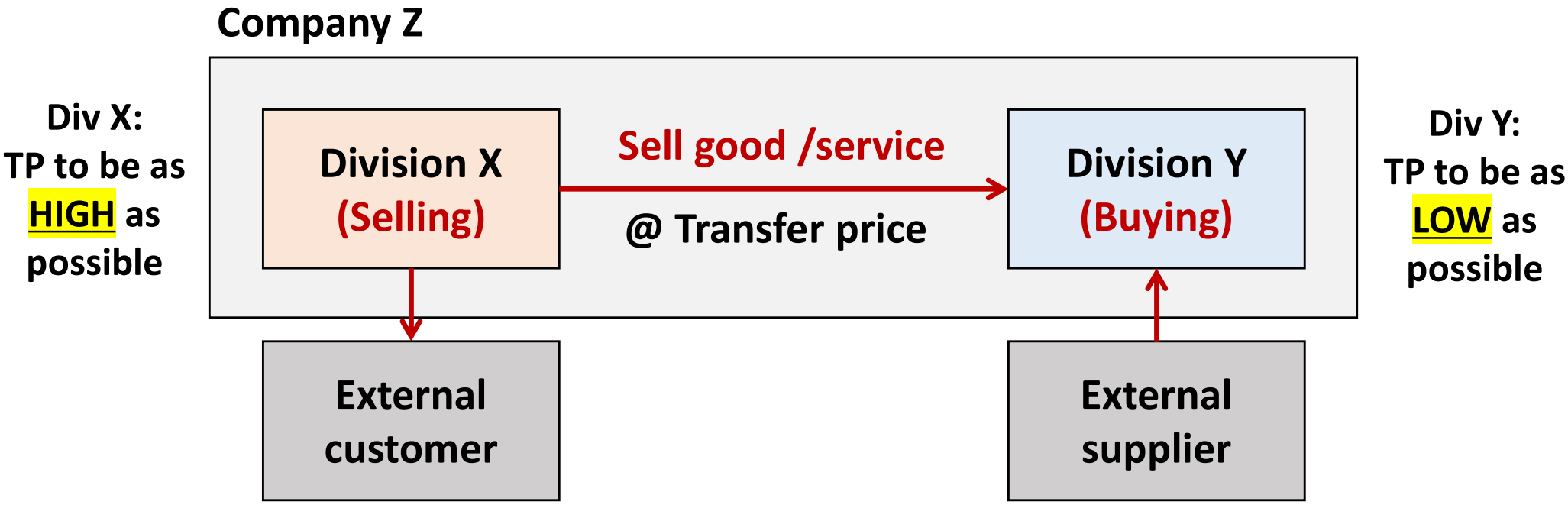
Learning Objective

Understand the impact of transfer pricing on a business

- Methods of determining transfer price
- How transfer pricing influences behavior that results in suboptimal decisions

Transfer Pricing

When division managers are given autonomy to decide whether to transfer internally or transact with external parties, they may make **suboptimal decisions** – their decisions may maximize divisional profit but not profit of the company as a whole.



Best Interest of Company:
Transfer internally if this will INCREASE Company's Profit.

Methods of Setting Transfer Price

Common approaches to set transfer prices:

- **Negotiated**: Leave it to divisions to set their own transfer price. Divisions would want transfer price that maximizes their own profits.
- **Market-based**: Based on price charged for an item on the open market.
- **Cost-based**: Based on cost of the selling division (e.g. variable cost, full absorption cost incurred). Usually cost plus % of markup.

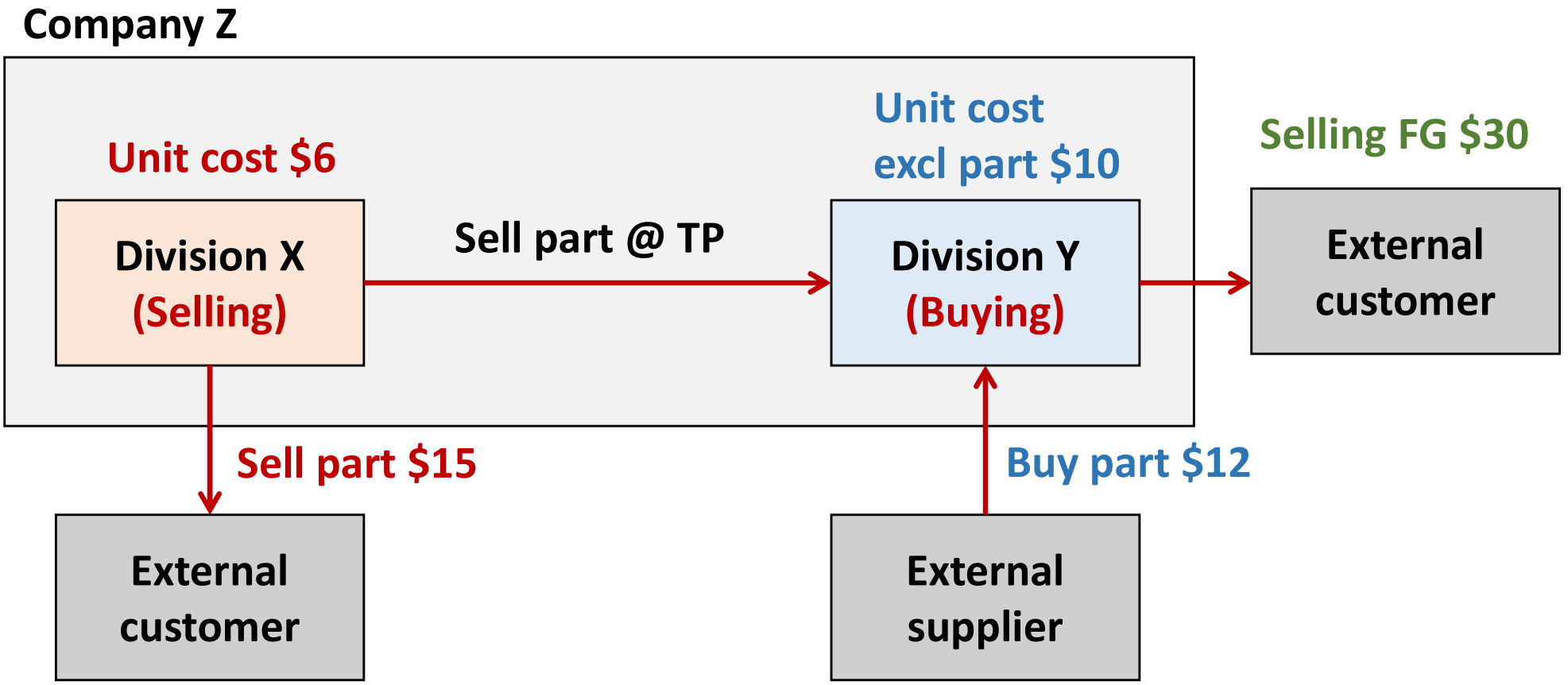
None is the BEST approach!
None can ensure TP will guarantee Divisions' interests are aligned with Company's interests.

Transfer Pricing – Example

Company Z has 2 divisions: Div X and Div Y.

- Div X produces a part that it can sell to externally at $15 or internally to Div Y.
- The cost of producing the part by Div X is $6. The entire cost is avoidable.
- Div Y produces a finished product that uses the part.
- Div Y can buy the part externally for $12 or internally from Div X.
- The cost of producing the finished product excluding the cost of the part = $10.
- Div Y sells the finished product at $30.

Transfer Pricing – Example



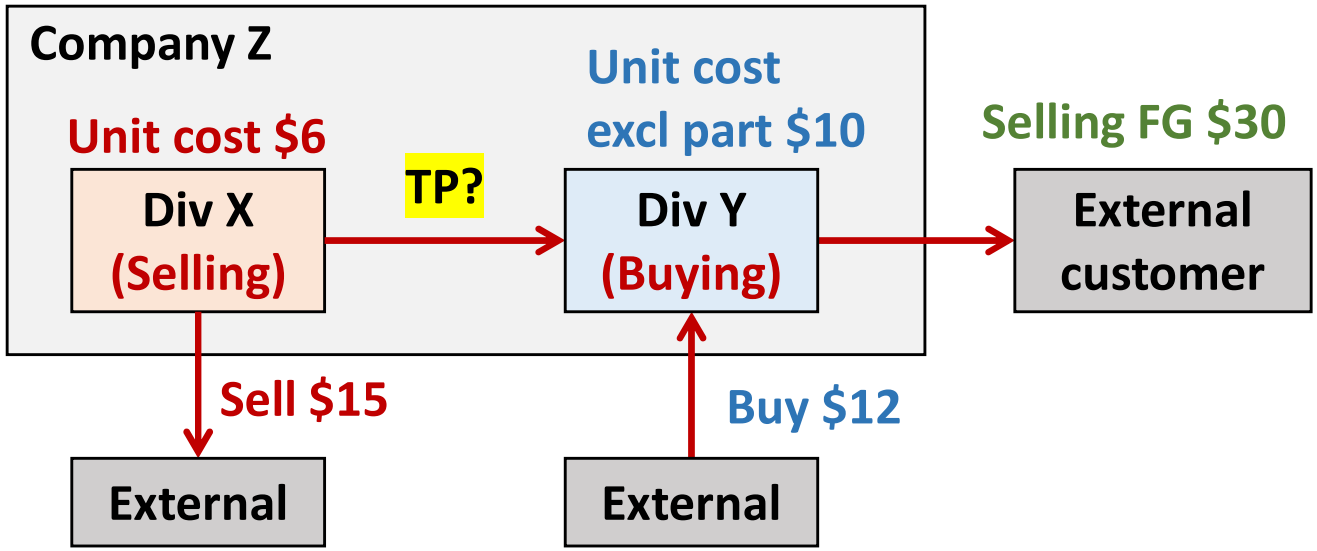
Division X:

1. No excess capacity (either External or Div Y but not both)
2. Excess capacity

Division Y:

1. Can buy from external supplier
2. Exhausted all external supply

Div X: No excess capacity (either External or Div Y but not both)



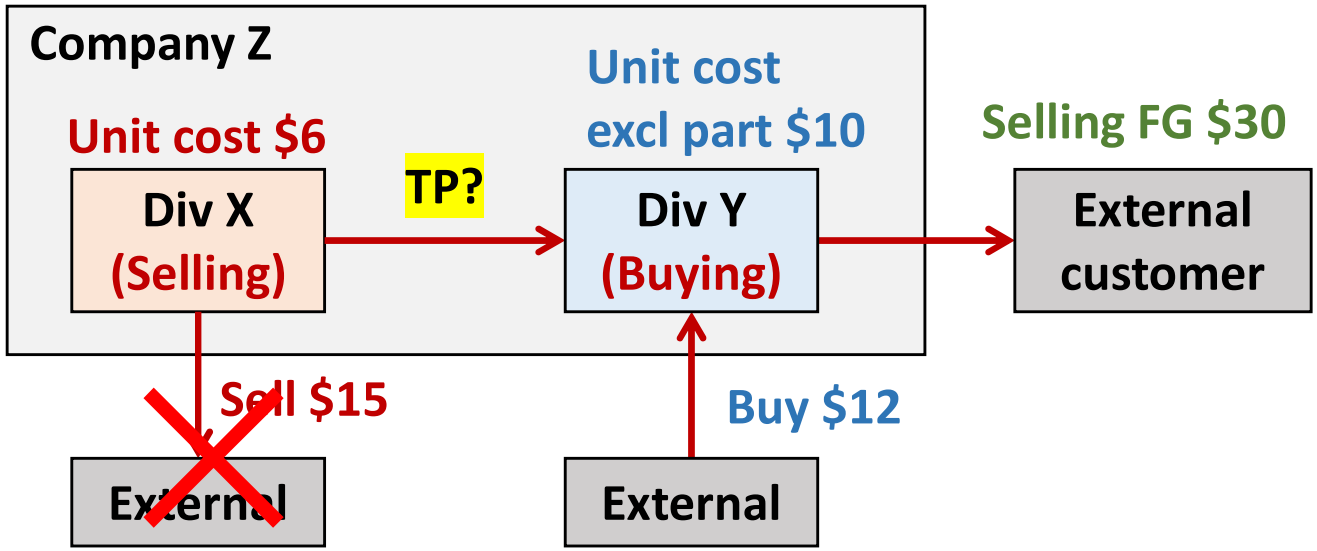
Div X has to produce the part and incur $6 regardless of whom it is selling it to. Cost of producing the part is not a relevant cost.

Div X

Opportunity cost if sell to Div Y =$15

TP ≥ $15

Div X: Excess capacity



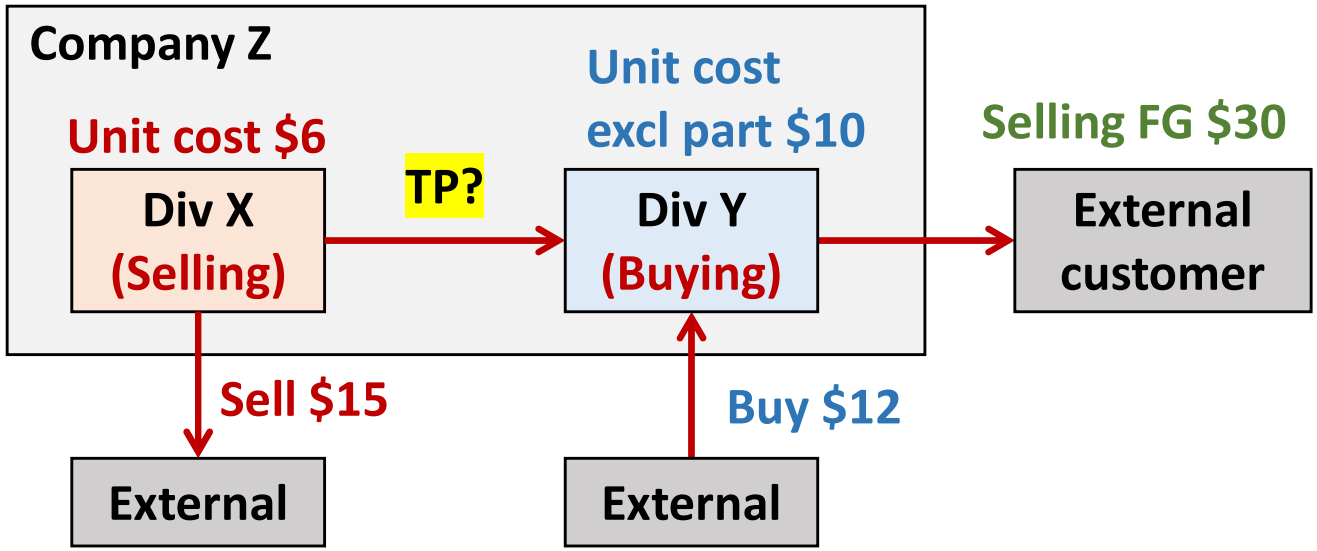
Div X

No opportunity cost if sell to Div Y

Relevant cost = Incremental cost = $6

TP ≥ $6

Div Y: Can buy from external supplier

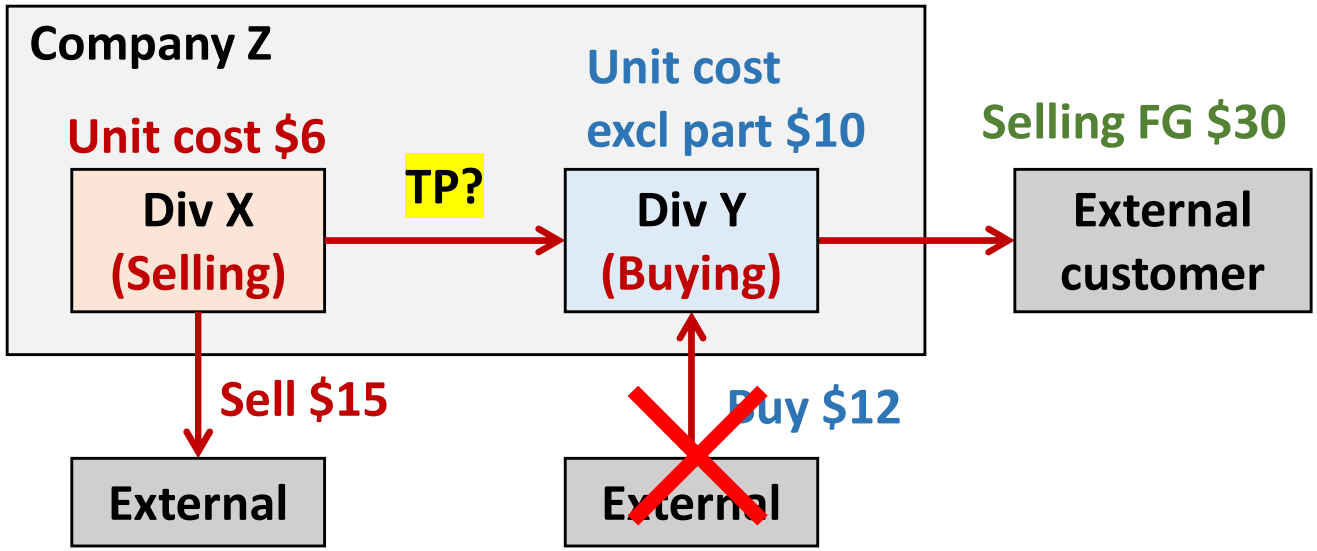


Div Y

Relevant cost = $12

TP ≤ $12

Div Y: Exhausted all external supply

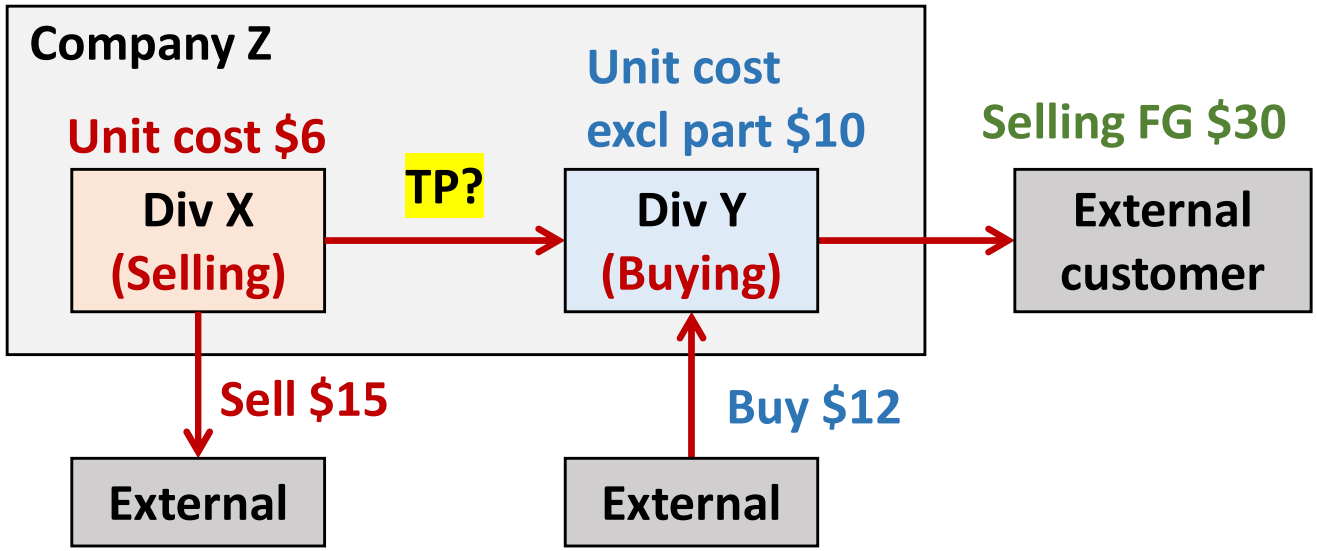


Div Y

Max cost of part = $30 - $10 = $20

TP ≤ $20

Div Y: Exhausted all external supply



Division X:

1. No excess capacity (either External or Div Y but not both)
2. Excess capacity

TP ≥ $15

TP ≥ $6

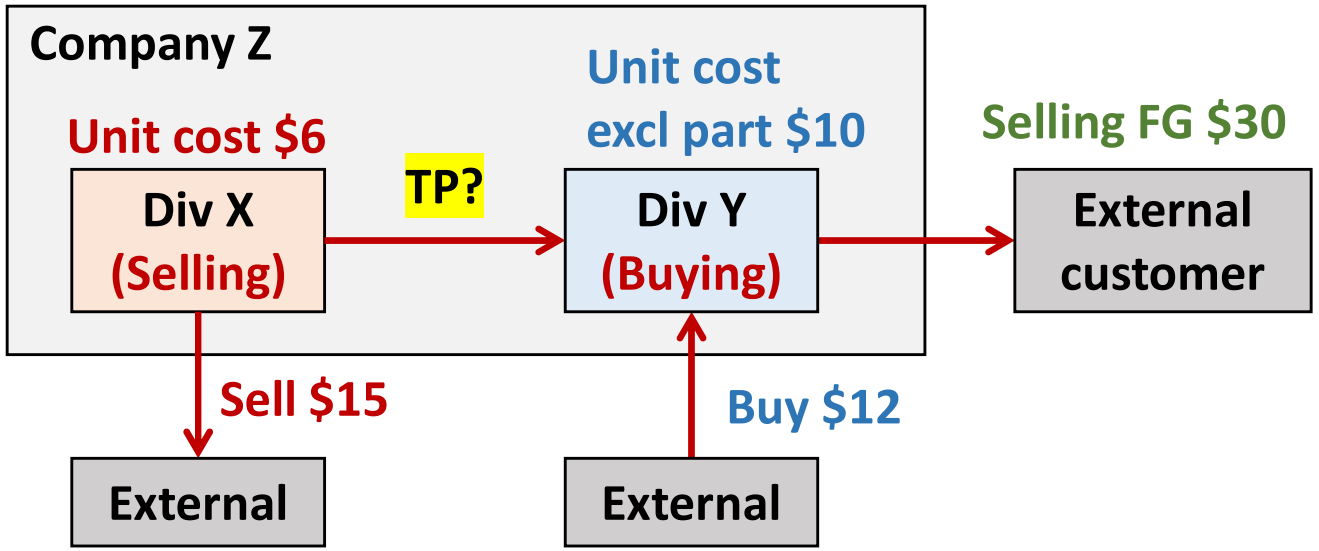
Division Y:

1. Can buy from external supplier
2. Exhausted all external supply

TP ≤ $12

TP ≤ $20

Negotiated TP (1)



Div X has to produce the part and incur $6 regardless of whom it is selling it to. Cost of producing the part is not a relevant cost.

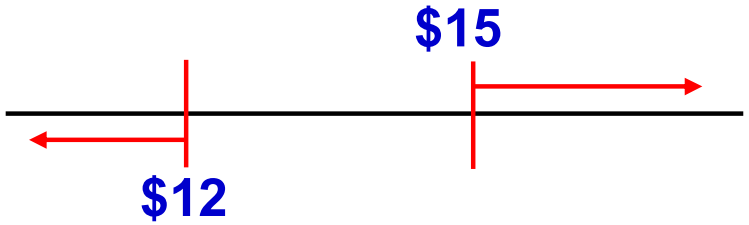
Div X does not have excess capacity. **Div Y can buy from outside supplier.**

Div X

Opportunity cost if sell to Div Y =$15
TP ≥ $15

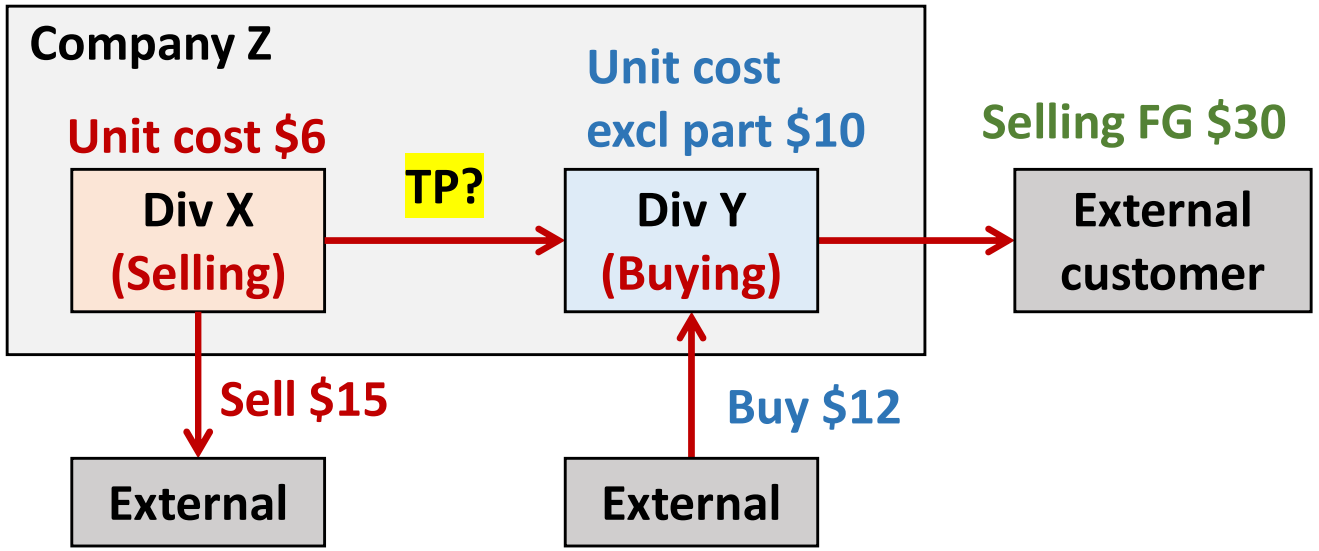
Div Y

Relevant cost = $12
TP ≤ $12



No internal transfer!

Negotiated TP (2)



Div X has to produce the part and incur $6 regardless of whom it is selling it to. Cost of producing the part is not a relevant cost.

Div X does not have excess capacity. **Div Y has exhausted all external supply.**

Div X

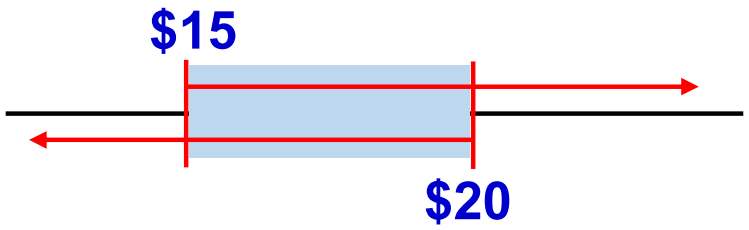
Opportunity cost if sell to Div Y =$15

TP ≥ $15

Div Y

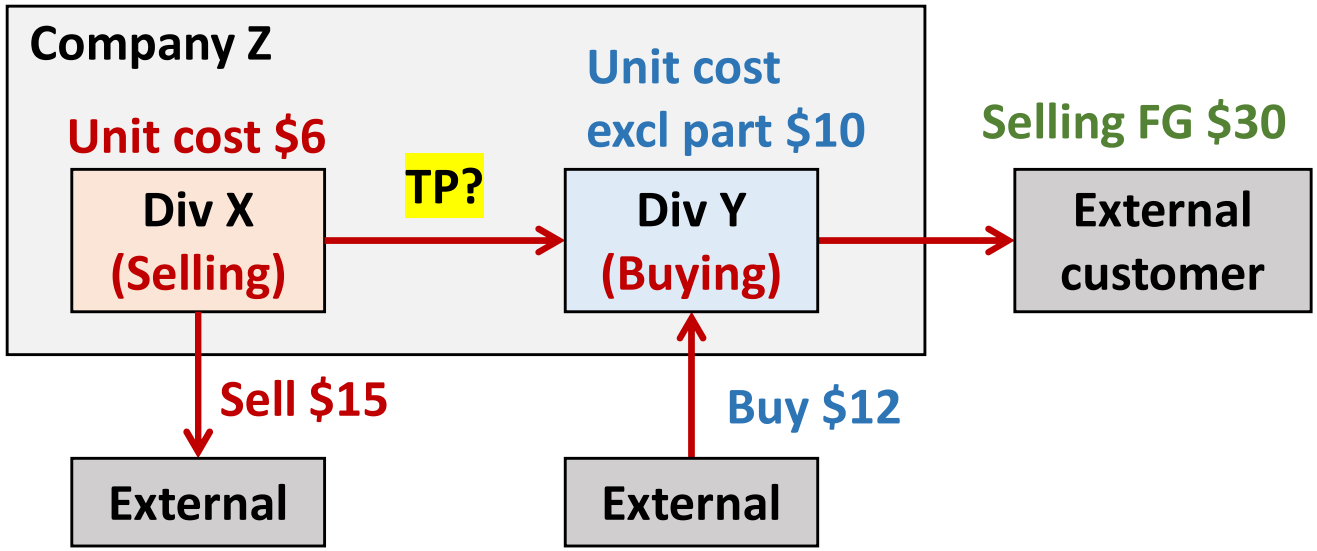
Max cost of part = $30 - $10 = $20

TP ≤ $20



$15 ≤ TP ≤ $20

Negotiated TP (3)



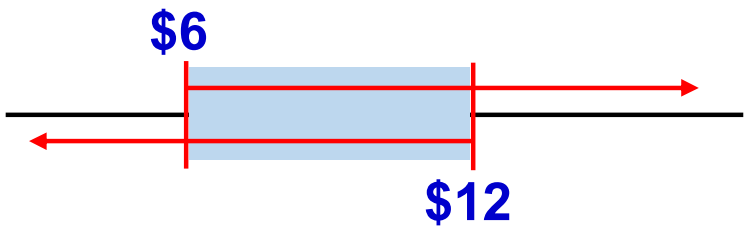
Div X have excess capacity. **Div Y can buy from outside supplier.**

Div X

No opportunity cost if sell to Div Y
Relevant cost = Incremental cost = $6
TP ≥ $6

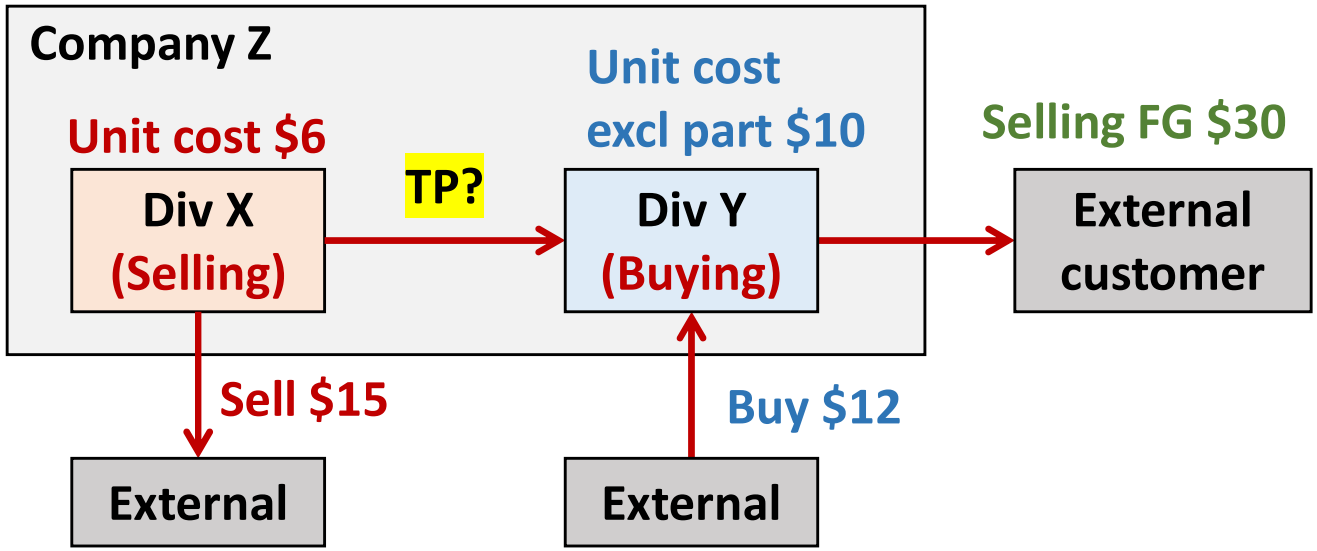
Div Y

Relevant cost = $12
TP ≤ $12



$6 ≤ TP ≤ $12

Negotiated TP (4)



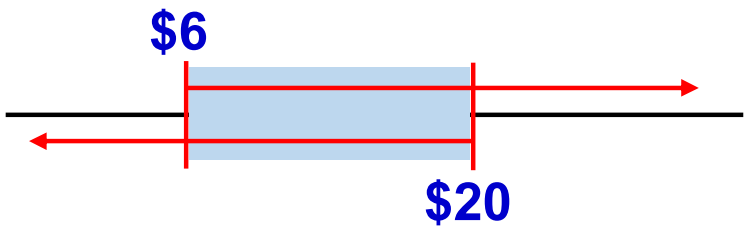
Div X have excess capacity. **Div Y has exhausted all external supply.**

Div X

No opportunity cost if sell to Div Y
Relevant cost = Incremental cost = $6
TP ≥ $6

Div Y

Max cost of part = $30 - $10 = $20
TP ≤ $20

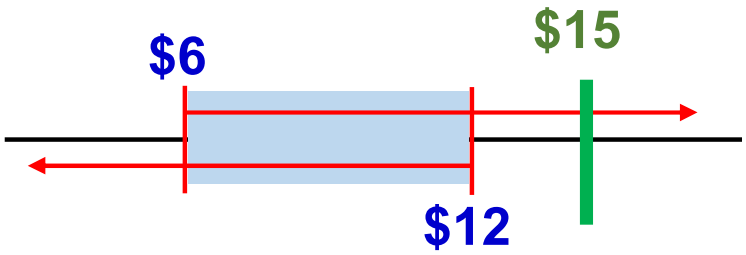


$6 ≤ TP ≤ $20

Transfer Pricing

- ***What if Div X is greedy and insists on TP of $15?***
- ***What if Company Z implements a market-based TP of $15?***

Div X have excess capacity. **Div Y can buy from outside supplier.**



$6 ≤ TP ≤ $12

Div Y will not buy internally *as it is cheaper to buy externally at $12.*

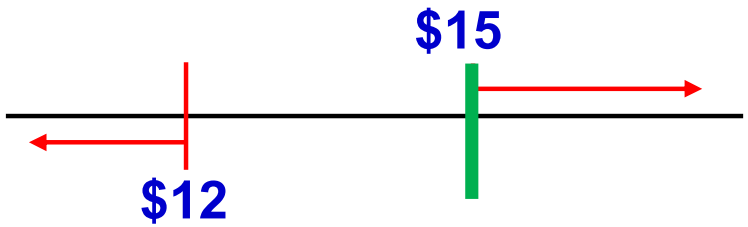
No transfer	**Relevant cost – External price for Div Y**	**$12**
Internal transfer	**Relevant cost – Div X cost of producing**	**$6**
Net disadvantage to whole company		**($6)**

Suboptimal decision!

Transfer Pricing

- ***What if Div X is greedy and insists on TP of $15?***
- ***What if Company Z implements a market-based TP of $15?***

Div X does not have excess capacity. **Div Y can buy from outside supplier.**



No internal transfer!

Div Y will not buy internally *as it is cheaper to buy externally at $12.*

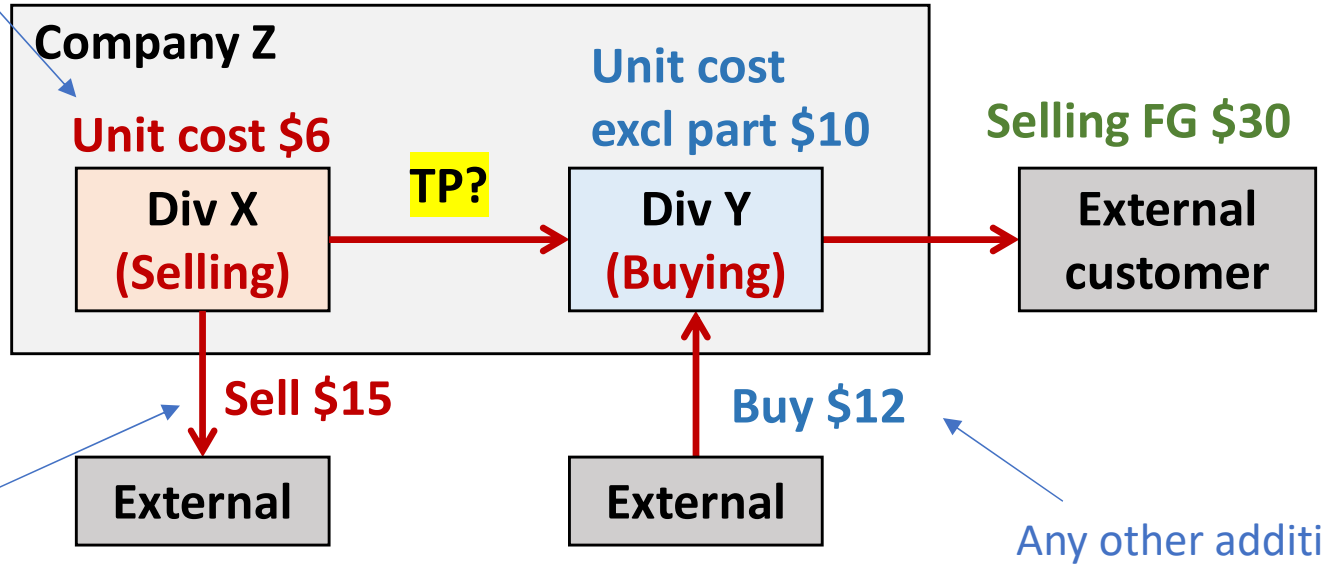
No transfer	**Relevant cost – External price for Div Y**	**$12**
Internal transfer	**Relevant cost – Opp cost of Div X revenue**	**$15**
Net advantage to whole company		**$3**

Good for Company Z!

Div X has to produce the part and incur $6 regardless of whom it is selling it to.
Cost of producing the part is not a relevant cost.

Things that may affect TP

Is there unavoidable fixed cost?
If so, not the whole $6 may be relevant.



Any additional selling cost?
This will decrease the CM
from selling externally.

Any other additional cost?
This will increase the total
cost of buying externally

Evaluation of Transfer Pricing methods

Negotiated TP

- Decentralization and preservation of the autonomy of the divisions.
- Managers negotiated have better information on costs on benefits.
- Pitting managers against each other may create a non-cooperative atmosphere.
- Heavily dependent on manager's negotiations skills.

Market-based TP

- Often work best if the there is an external demand for such products.
- Will not work well if selling division has idle capacity.

Cost-based TP

- TP based on full cost (including fixed cost) will lead to suboptimal decisions
- Cost-based prices do not provide incentives to control cost since the cost is just going to be passed on to the buying division.

When Top Management Decide

- No one perfect method to set transfer price.
- Should Top Management intervene when Divisions' decision results in suboptimal decision for the company as a whole?

Intervention defeats purpose of decentralization.

- For tax advantages

Division X (Selling) in Country A	Division Y (Buying) in Country B
Tax Rate = 15%	Tax Rate = 35%

  - Top Management will want TP to be as high as possible – Lower profits in countries with higher tax rates.
  - However, less and less relevant. Tax authorities are not "stupid".
    - TP to be same as an arms length transactions
    - Global minimum effective tax rate

End of topic